

TrES-3b

R-0312

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_170527 · created 2026-07-17T23:13:08+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457900.7071 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.055
Transit depth (Rp/Rs)^2	3.38 +/- 2.04 [%]
Orbital Inclination (inc)	80.79 +/- 2.82
Airmass coefficient 1 (a1)	1.026 +/- 0.095
Airmass coefficient 2 (a2)	-0.005 +/- 0.04
Scatter in the residuals of the lightcurve fit is	9.32 %
Best Comparison Star	#2 - [314, 288]
Optimal Method	PSF photometry
Transit Duration (day)	0.04 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.184 ± 0.055 vs 0.16309 (deviation 0.38σ)
Duration fit vs published	0.04 d vs 0.0567 d (deviation 0.56σ)
Mid-time O–C	2.4 min
Depth SNR	3.3
Residual scatter	9.32 %

Light curve

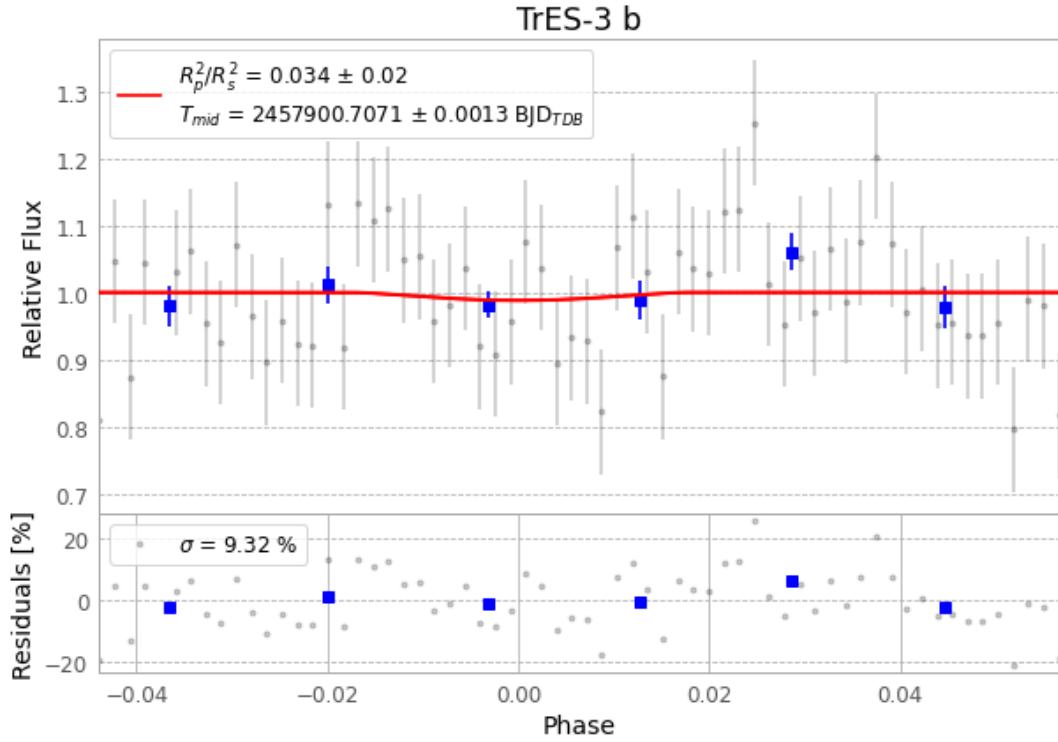


Figure 1 — FinalLightCurve_TrES-3 b_2017-05-26.png (SHA-256 0715bf02b8e0f58f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [224, 184], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f786f14d33dce36f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

64 FITS frames (42 MB), TRES-3170527033012.FITS → TRES-3170527063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-170527.log (546b6a9624e0...), FinalLightCurve_TrES-3 b_2017-05-26.png (0715bf02b8e0...), FinalLightCurve_TrES-3 b_2017-05-26.csv (ba3900d35a17...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 23:13Z.